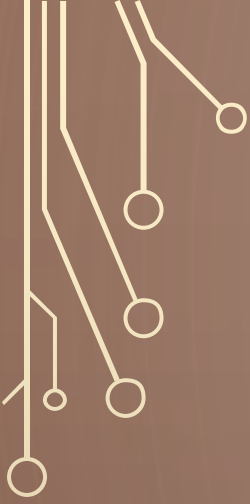




HABOTCONNECT HIRING PROJECT

SECURE CLOUD INFRASTRUCTURE PROVISIONING
USING TERRAFORM, GOOGLE CLOUD PLATFORM,
GITHUB ACTIONS & DJANGO REST FRAMEWORK

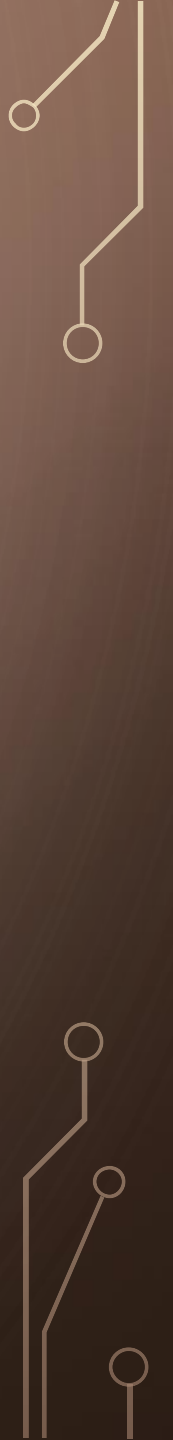
AMRUTA SURYAVANSHI



PROJECT OVERVIEW

- • Provision secure GCP staging infrastructure
- Automate deployment using Terraform
- Protect student data
- Implement Fail-Closed CI/CD

PROJECT OBJECTIVES

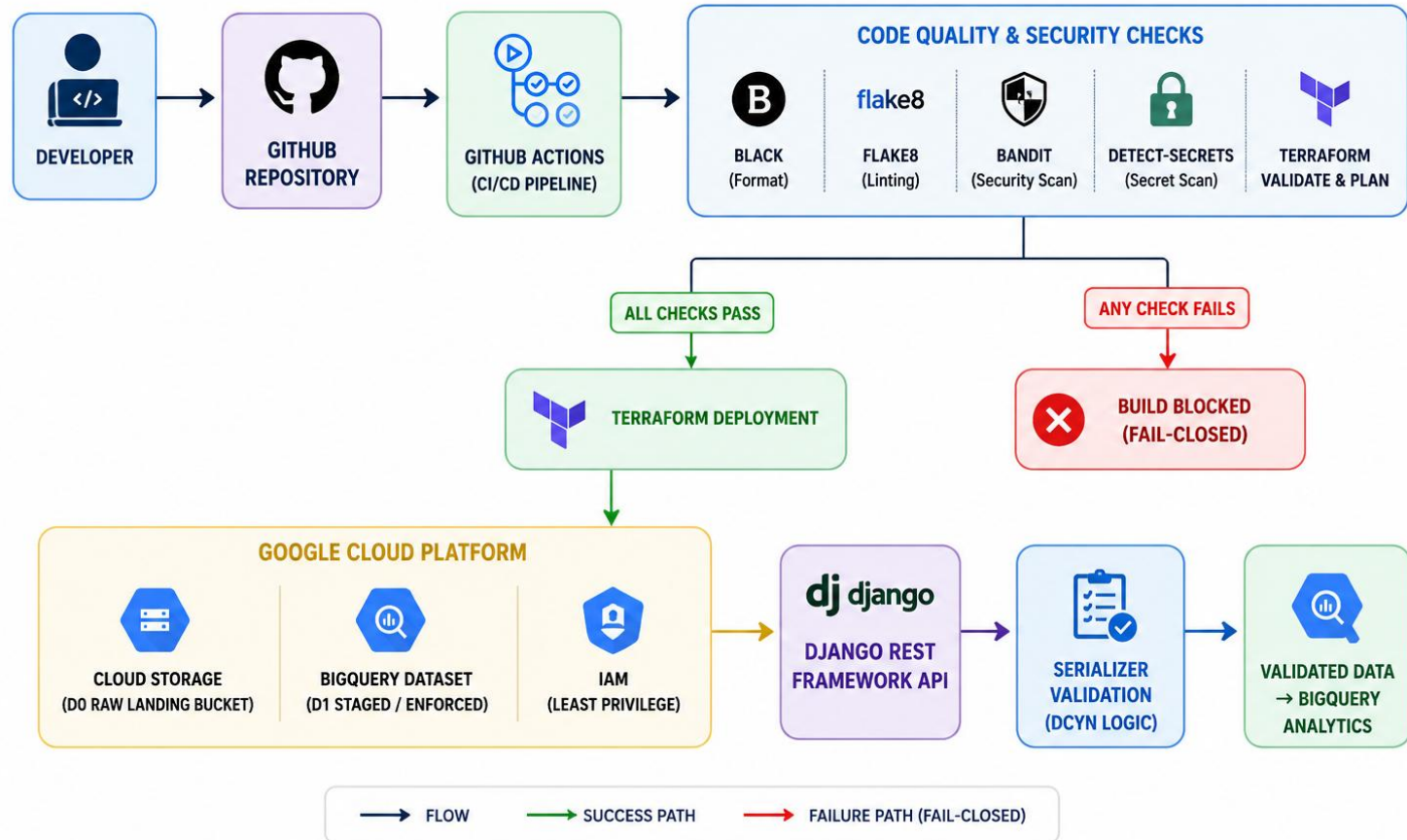
- Infrastructure as Code
 - Secure IAM & Least Privilege
 - Row-Level Security
 - Automated validation
- 
- 

TECHNOLOGY STACK

- Terraform
 - Google Cloud Platform
 - GitHub Actions
 - Django REST Framework
 - BigQuery
 - Cloud Storage

PROJECT ARCHITECTURE

PROJECT ARCHITECTURE



TASK 1: TERRAFORM

- • Provider & Variables
- GCS Raw Landing Bucket
- BigQuery Dataset
- IAM
- Public Access Prevention
- Versioning

SECURITY IMPLEMENTATION

- • IAM (RBAC)
 - Principle of Least Privilege
 - Row-Level Security (RLS)
 - Secret Protection

TASK 2: CI/CD

- Pipeline:

Checkout → terraform fmt → validate → flake8 →
Secret Scan → Deploy

Successful pipeline

The screenshot displays the GitHub Actions interface for a repository named 'SuryavanshiAmruta / habot-devops-project'. The 'Actions' tab is selected, showing a workflow named 'Habot Secure CI Pipeline' with a job 'updated main.tf #39' that has completed successfully. The interface includes a sidebar with navigation options like 'Summary', 'All jobs', 'security-checks', 'Run details', 'Usage', and 'Workflow file'. The main content area shows the job's status as 'Success' with a total duration of '23s'. Below this, a section titled 'ci.yml on: push' shows a step named 'security-checks' that completed successfully in '14s'.

Summary

security-checks

Run details

Usage

Workflow file

Triggered via push now

Triggered via	Status	Total duration	Artifacts
SuryavanshiAmruta pushed -> 606b0f7 main	Success	23s	-

ci.yml
on: push

security-checks 14s

FAIL-CLOSED BUILD GATE

- Pipeline immediately stops if:
 - Formatting fails
 - Linting fails
 - Hardcoded API key detected
 - Terraform validation fails

TASK 3: DRF VALIDATION

- • Student onboarding JSON
 - Serializer validation
 - DCYN Yes/No logic
 - Strict field validation

TESTING & RESULTS

- Terraform Apply Successful
- Bucket & Dataset Created
- IAM Verified
- Pipeline Tested
- API Validation Successful

Terraform Apply

```

google_bigquery_dataset.staging: Creation complete after 3s [id=projects/habot-devops-project-504219/datasets/d1_staged]
google_storage_bucket.raw_landing: Creation complete after 4s [id=habot-raw-landing-504219]
google_storage_bucket_iam_member.viewer: Creating...
google_storage_bucket_iam_member.viewer: Creation complete after 6s [id=b/habot-raw-landing-504219/roles/storage.objectViewer/user:amrutasuryavanshi212@gmail.com]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

bucket_name = "habot-raw-landing-504219"
dataset_name = "d1_staged"
PS D:\Devops_projects\habot-devops-project\terraform>

```

Storage bucket

Buckets

Create

Refresh

Go to path

Learn

Filter

Filter buckets

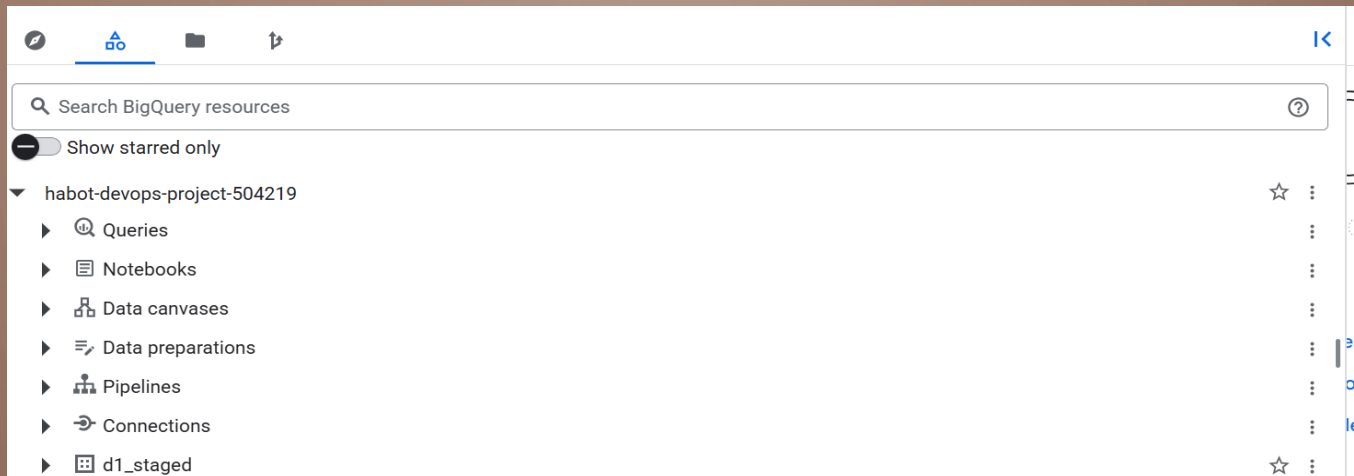
?

⚙

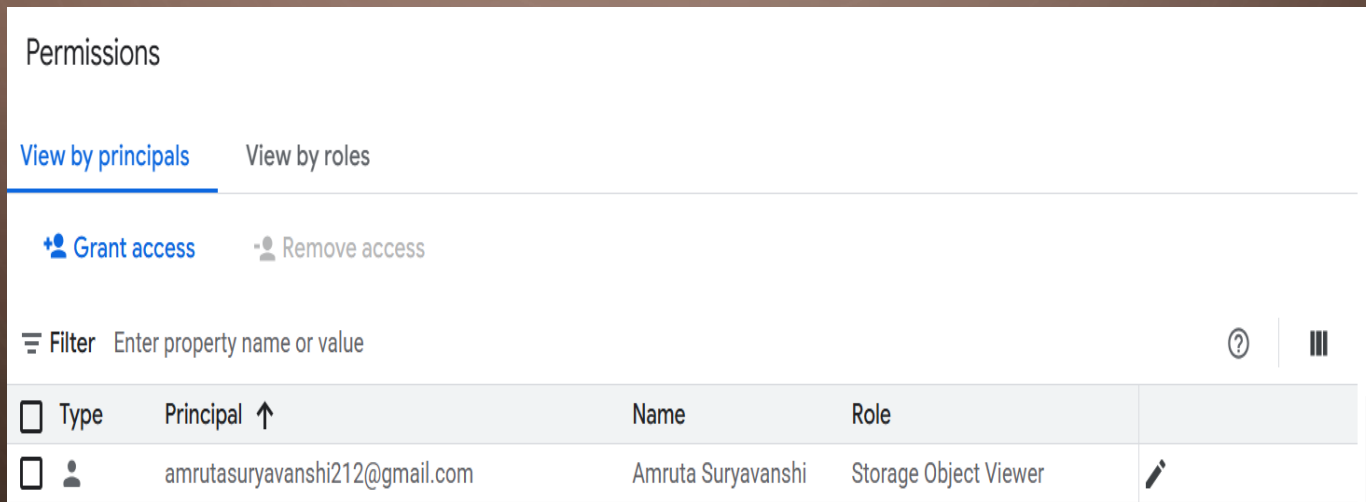
☰

Name ↑	Created	Location type	Location	Default storage class ?	Last modified
habot-raw-landing-504219	Aug 2, 2026, 1:41:59 AM	Multi-region	asia	Standard	Aug 2, 2026, 1:41:59 AM

BigQuery dataset



IAM Verified



Pipeline Tested

The screenshot shows the GitHub Actions interface for a workflow named 'Habot Secure CI Pipeline'. The workflow is triggered by a push to the 'main' branch. The status is 'Success' with a total duration of 23s. The workflow file is 'ci.yml' and the job is 'security-checks', which completed successfully in 14s.

Summary

All jobs

- security-checks

Run details

- Usage
- Workflow file

Triggered via push now

SuryavanshiAmruta pushed · 606b0f7 · main

Status: Success

Total duration: 23s

Artifacts: -

ci.yml

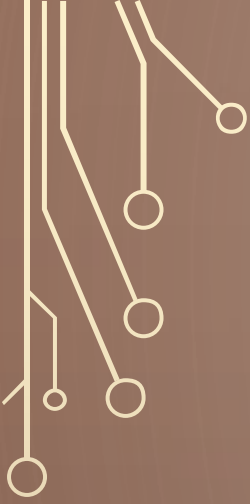
on: push

security-checks 14s

API Validation Successful

The screenshot shows the Databricks Data Catalog interface for the 'student_data' table. The table is located in the 'habot-devops-project-504219' workspace, under the 'd1_staged' dataset. The schema is displayed in the 'Schema' tab, showing the following fields:

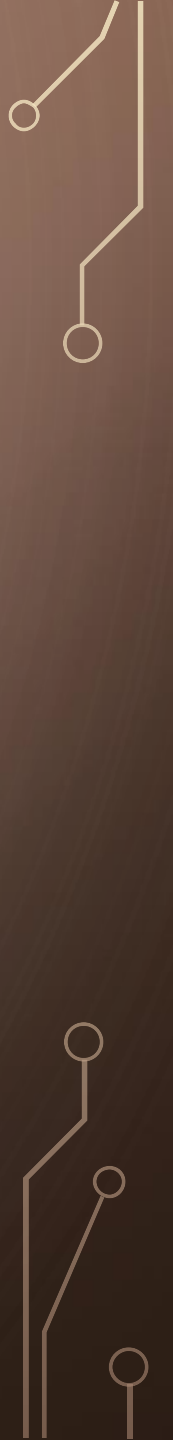
Field name	Type	Mode	Description	Key	Collation	Default Value	Dist
student_name	STRING	REQUIRED	-	-	-	-	-
school	STRING	REQUIRED	-	-	-	-	-
city	STRING	NULLABLE	-	-	-	-	-
age	INTEGER	NULLABLE	-	-	-	-	-



CHALLENGES

- Learning IAM & RLS
- CI/CD integration
- Cloud security

FUTURE IMPROVEMENTS

- Remote Terraform state
 - Secret Manager
 - Cloud Monitoring
 - Multi-environment deployment
- 
- 

CONCLUSION

- Project meets HaborConnect requirements using secure IaC, automated CI/CD, robust validation, and cloud security best practices.

The background is a dark brown gradient with faint, concentric circles centered around the text. In the corners, there are stylized, light-colored lines resembling circuit traces or a network diagram, with small circles at various points.

THANK YOU